Assignment No 5

Name :- Katkar Abhijit **Div:-** B

Batch:-T3 **Roll No :-** CO3B18

Title – Use BFS to solve a planning problem. Examples can be 8 puzzle problem, Robot path planning,Blocks world etc.

Program:-

```
from collections import deque


def bfs(grid, start, goal):
    rows = len(grid)
    cols = len(grid[0])

    queue = deque([start])
    visited = {start}
    parent = {start: None}

    # Up, Down, Left, Right
    directions = [
        (-1, 0),
        (1, 0),
        (0, -1),
        (0, 1)
    ]
```

```
while queue:
    current = queue.popleft()

    if current == goal:
        break

    for dr, dc in directions:
        nr = current[0] + dr
        nc = current[1] + dc
        next_position = (nr, nc)

        if (0 <= nr < rows and
            0 <= nc < cols and
            grid[nr][nc] != 1 and
            next_position not in visited):

            visited.add(next_position)
            parent[next_position] = current
            queue.append(next_position)

# If goal was not reached
if goal not in parent:
```

```
        return None

    # Create path
    path = []
    current = goal

    while current is not None:
        path.append(current)
        current = parent[current]

    path.reverse()

    return path



# 0 = free path
# 1 = obstacle

grid = [
    [0, 0, 0, 1, 0],
    [1, 0, 0, 1, 0],
    [0, 0, 1, 0, 0],
    [0, 0, 0, 0, 0]
```

```
]

start = (0, 0)
goal = (3, 4)

path = bfs(grid, start, goal)

if path:
    print("Shortest Path:")
    print(path)
    print("Number of moves:", len(path) - 1)
else:
    print("No path found")
```

Output:-

```
Shortest Path:
[(0, 0), (0, 1), (1, 1), (2, 1), (3, 1), (3, 2), (3, 3), (3, 4)]
Number of moves: 7

Process finished with exit code 0
```